Jerico Agdan
November 2025

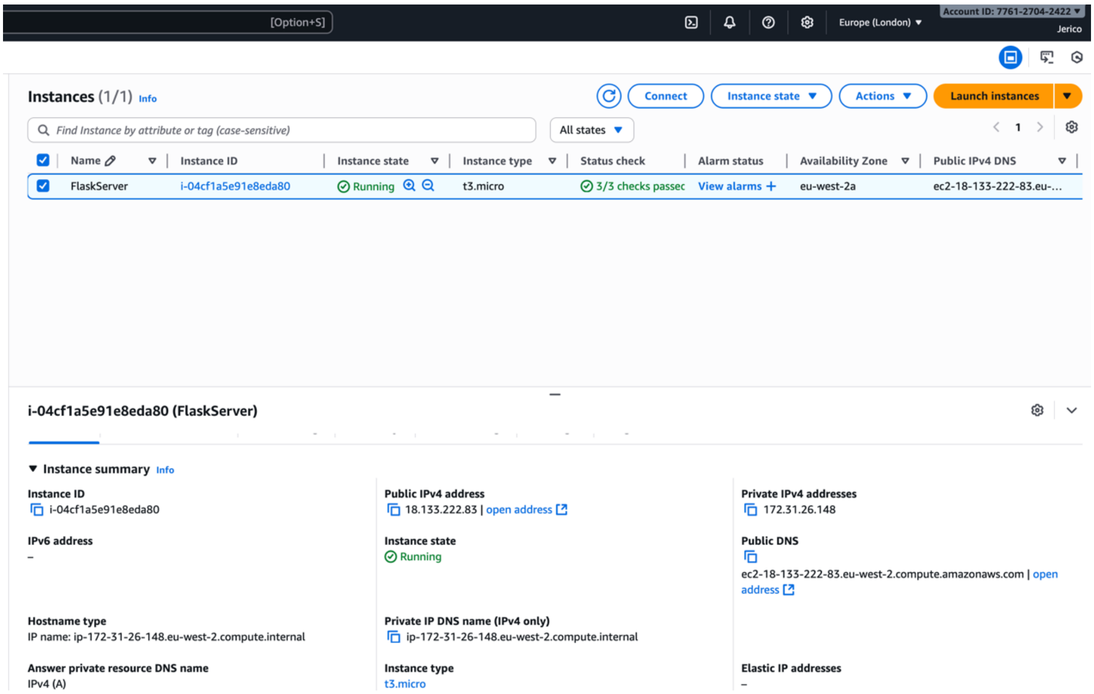
Exercise 1

You can find the Flask code and HTML in my GitHub account (https://github.com/JericoNeil/Cloud-Computing-Flask-Assignment.git). The **Read Me** file explain step by step how I created the API about the language detection. It is available specifically in the "**Assignment1_LanguageDetect"** folder.

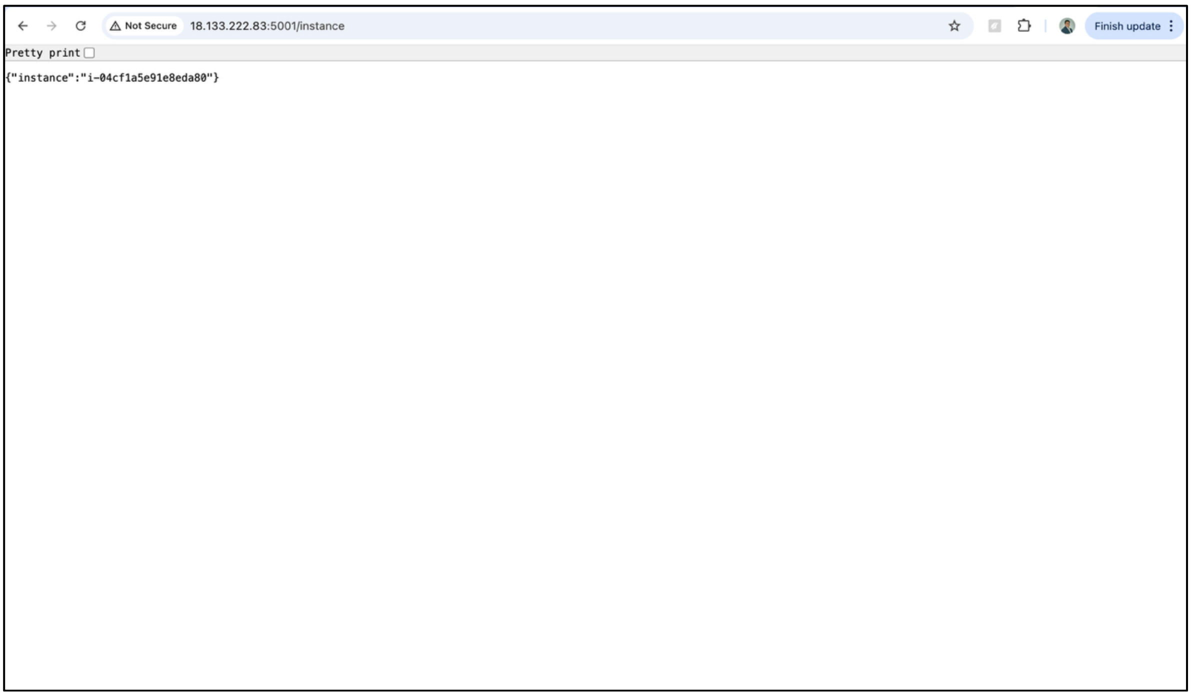
Exercise 2

1) Screenshot of the running instance

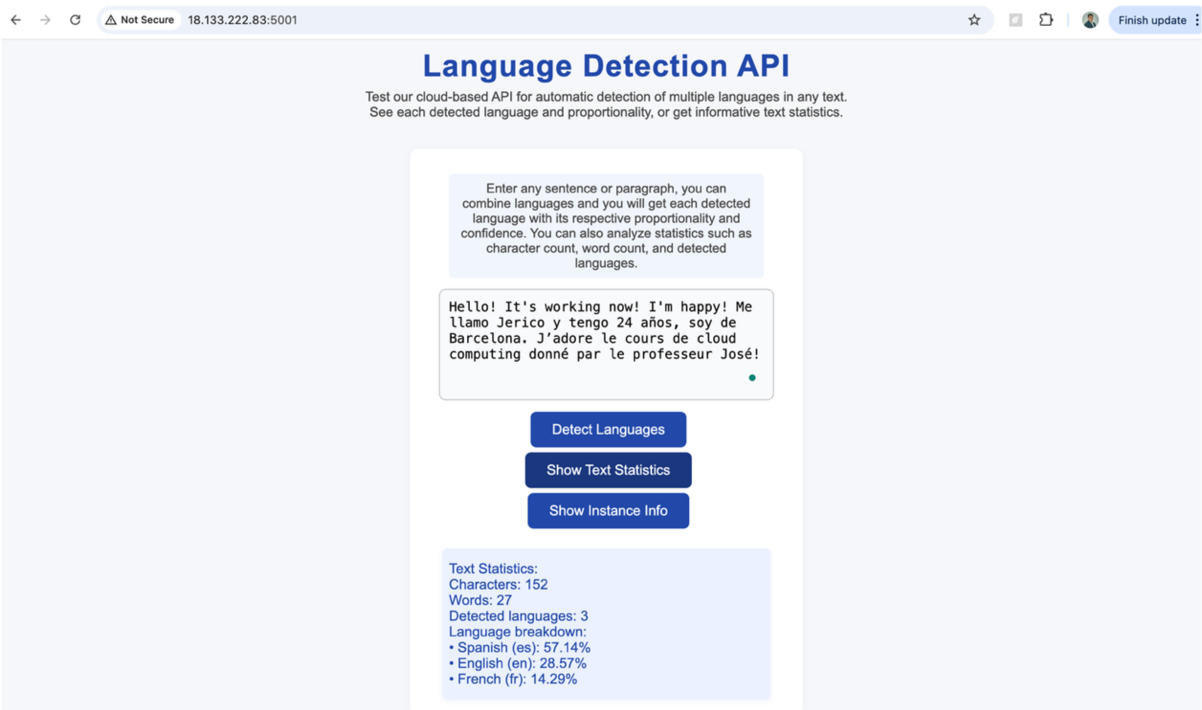
Screenshot of AWS's instances with the list of instances, with the instance in "Running" state, the public IP and username at the top right:



2) **Screenshot of the request to the /instance endpoint**



3) Extra: Screenshot of the API working with the Text Statistics endpoint



It shows the number of characters, words, detected languages and language breakdown with its respective proportion

Exercise 3

You can find this exercise in my GitHub account: https://github.com/JericoNeil/Cloud-Computing-Flask-Assignment.git

The **Read Me** file explains more about the project. All materials for this exercise are also located in my GitHub account, specifically in the "**Assignment3_CareerPath**" folder.

1. INTRODUCTION OF THE PROBLEM

For this exercise, I developed an advisory application that enables high school students to input their Bachillerato and PAU grades and discover a tailored set of university degrees they may realistically pursue based on their predicted or actual scores.

The motivation behind this project is personal: I want to help my brother decide which university degree to aim for. He is currently in his first year of high school and, like many students, is unsure about what field to choose. Despite the significance of this decision, there is no existing platform that shows students the full spectrum of career paths available based on their predicted or actual high school and PAU grades. This project is designed to provide actionable and grade-driven insights to guide students toward informed decisions about their futures.

In Catalonia, you can find the website from the Government of Catalonia about the cut-off grades of Catalonian bachelor's, but the information of studies is in PDF, which is not very convenient to filter degrees: https://universitats.gencat.cat/ca/preinscripcions/notes-tall/

Generalitat de Catalunya
Consell Interuniversitari de Catalunya
Oficina d'Accés a la Universitat

Notes de tall 1a assignació Juny 2025 (11/07/25)						
Ordre alfabètic per centre d'estudi						
ESTUDIS DE GRAU I TÍTOLS PROPIS						
Codi	*Nom del centre d'estudi i població*	*Sigles Universitat*	*PAU / CFGS*	*Més grans de 25 anys*	*Titulats universitaris*	*Més grans de 45 anys*
41029	Administració d'Empreses i Gestió de la Innovació "Tecnocampus" (Mataró)	UPF	7,688	5,000	5,000	5,000
41036	Administració d'Empreses i Gestió de la Innovació "Tecnocampus" (docència en Anglès) (Mataró)	UPF	7,628	5,000	5,000	5,000
41041	Administració d'Empreses i Gestió de la Innovació / Màrqueting i Comunitats Digitals "Tecnocampus" (simultaneïtat) (Mataró)	UPF	7,546	5,000	5,000	5,000
41035	Administració d'Empreses i Gestió de la Innovació / Turisme i Gestió del Lleure "Tecnocampus" (simultaneïtat) (Mataró)	UPF	5,000	5,000	5,000	5,000
11033	Administració i Direcció d'Empreses (Barcelona)	UB	9,348	5,000	5,000	5,000
41008	Administració i Direcció d'Empreses (Barcelona)	UPF	10,966	5,000	5,000	5,000
71063	Administració i Direcció d'Empreses "IGEMA" (Barcelona)	URV	5,000	5,000	5,000	5,000
21014	Administració i Direcció d'Empreses (Cerdanyola del Vallès)	UAB	9,804	5,000	5,000	5,000
21099	Administració i Direcció d'Empreses (docència en Anglès) (Cerdanyola del Vallès)	UAB	10,128	5,000	5,000	5,000
81003	Administració i Direcció d'Empreses (Girona)	UdG	9,218	5,000	5,000	6,333
61069	Administració i Direcció d'Empreses (Igualada)	UdL	8,224	5,000	5,000	5,000
61003	Administració i Direcció d'Empreses (Lleida)	UdL	7,764	5,000	5,000	5,000
91042	Administració i Direcció d'Empreses (Campus Manresa) (Manresa)	UVic-UCC	5,716	5,000	5,000	5,000
71002	Administració i Direcció d'Empreses (Reus)	URV	8,413	5,000	5,000	5,000
71004	Administració i Direcció d'Empreses (Tortosa)	URV	7,220	5,000	5,000	5,000
91003	Administració i Direcció d'Empreses (Vic)	UVic-UCC	6,420	5,000	5,000	5,000
11064	Administració i Direcció d'Empreses / Dret (simultaneïtat) (Barcelona)	UB	10,692	5,000	6,860	5,000

Moreover, you also have this platform from notasdecorte.es (https://notasdecorte.es/zona/cataluna), in which you can have access to all degrees not only from Catalonia. This tool offers filters to select a specific university or to search for a particular degree if the student already knows what they are looking for. However, many students especially those like my brother who are still exploring their options, may not yet have a clear target in mind and would benefit from a broader, more exploratory tool.

Notas de corte: Cataluña 2024-2025

La nota de corte más alta de Cataluña en el curso 2024-2025 fue de un 13,111 para estudiar el Doble Grado en Física + Matemáticas en la Universitat Autònoma de Barcelona

Abajo se muestran datos de los 856 grados ofrecidos en Cataluña, de los que 506 se imparten en centros públicos, y 350 en centros privados

Importante: **Recuerda que es imposible saber de antemano qué nota de acceso tendrás que sacar para entrar en cualquier carrera de Cataluña este año.** Las notas de corte del año pasado son sólo orientativas, ya que cambian cada año en función de la demanda y del número de plazas ofrecidas

Filtrar por universidad

Barcelona Culinary Hub
Bau - Centro Universitario de Diseño de Barcelona
CESDA (Centro de Estudios Superiores de la Aviación)
CETT - Campus de Turismo, Hotelería y Gastronomía
Campus Docent Sant Joan de Déu
Centro de la Imagen y la Tecnología Multimedia (CITM)
E.U. de Enfermería del Hospital de la Santa Creu i Santa Pau
E.U. de Hotelería y Turismo Sant Pol de Mar
E.U. de Relaciones Laborales de Lleida
E.U. de Turismo Euroaula
E.U. de Turismo y Dirección Hotelera
E.U. de la Salud y el Deporte (URV)
E.U. de la Salud y el Deporte (UdG)
EAE Business School
EINA, Centro Universitario de Diseño y Arte de Barcelona (UAB)
ELISAVA Escuela Universitaria de Diseño e Ingeniería de Barcelona

Titulaciones

Doble Grado en Física + Matemáticas
Universitat Autònoma de Barcelona
Universidad Pública
Web de la facultad: http://www.uab.cat/ciencies
Duración: 5,0 años
Precio del primer curso: 1.108 €
Pídeles información ¡GRATIS!

Barcelona
Presencial
Nota de corte
13,111
Idioma de enseñanza:
Bilingüe (castellano/lengua cooficial)

Doble Grado en Física + Matemáticas
Universitat de Barcelona
Universidad Pública
Web de la facultad: http://www.mat.ub.es
Duración: 5,0 años
Precio del primer curso: 1.181 €
Pídeles información ¡GRATIS!

Barcelona
Presencial
Nota de corte
12,864
Idioma de enseñanza:
Lengua cooficial

Hence, my project provides a more intuitive and user-focused experience by enabling students to discover degree programs that truly fit their academic profile and interests, even if they don't have a predetermined major or university in mind. The aim is to empower high school students to make more informed and personalized choices about their academic and career paths.

2. DATA CLEANING AND STRUCTURING

In order to carry out the project, I used the Database from the Generalitat of Catalonia website, which is the official website to retrieve the "cut-off grades" or "Notes de Tall" (in Catalan), of each degree of Catalonian universities (excluding private university bachelor's). Unfortunately, we do not have the Excel File, so **I converted the PDF file to Excel and followed a multi-step data cleaning process using Excel**.

Notes de tall 1a assignació Juny 2025 (11/07/25)						
Ordre alfabètic per centre d'estudi						
ESTUDIS DE GRAU I TÍTOLS PROPIS						
Codi	***Nom del centre d'estudi i població***	***Sigles Universitat***	***PAU / CFGS***	***Més grans de 25 anys***	***Titulats universitaris***	***Més grans de 45 anys***
41029	Administració d'Empreses i Gestió de la Innovació "Tecnocampus" (Mataró)	UPF	7.688	5.000	5.000	5.000
41036	Administració d'Empreses i Gestió de la Innovació "Tecnocampus" (docència en Anglès) (Mataró)	UPF	7.628	5.000	5.000	5.000
41041	Administració d'Empreses i Gestió de la Innovació / Màrqueting i Comunitats Digitals "Tecnocampus" (simultaneïtat) (Mataró)	UPF	7.546	5.000	5.000	5.000
41035	Administració d'Empreses i Gestió de la Innovació / Turisme i Gestió del Lleure "Tecnocampus" (simultaneïtat) (Mataró)	UPF	5.000	5.000	5.000	5.000
11033	Administració i Direcció d'Empreses (Barcelona)	UB	9.348	5.000	5.000	5.000
41008	Administració i Direcció d'Empreses (Barcelona)	UPF	10.966	5.000	5.000	5.000
71063	Administració i Direcció d'Empreses "IGEMA" (Barcelona)	URV	5.000	5.000	5.000	5.000
21014	Administració i Direcció d'Empreses (Cerdanyola del Vallès)	UAB	9.804	5.000	5.000	5.000
21099	Administració i Direcció d'Empreses (docència en Anglès) (Cerdanyola del Vallès)	UAB	10.128	5.000	5.000	5.000

Moreover, I also enriched the dataset by adding several **new columns (in red)** to be able to filter:

- **Location**: The city where each university program is offered
- **Province**: The province in which the university is located (for broader geographic filtering, there are 4 provinces in Catalonia)

- **Field**: The academic or professional field of each degree (such as Health Sciences, Engineering, etc.)
- **High-School Specialization**: The typical Spanish Bachillerato track granting access to each degree (e.g., Science and Technology, Social Sciences, Arts and Humanities)

Notes de tall 1a assignació Juny 2025 (11/07/25)							
Ordre alfabètic per centre d'estudi							
ESTUDIS DE GRAU I TÍTOLS PROPIS							
Code	Degree	University	Location	Province	Field	High-School Specialization	PAU
21169	Física / Matemàtiques (simultaneïtat) (Cerdanyola del Vallès)	UAB	Cerdanyola del Vallès	Barcelona	Double Degree	Science and Technology	13.370
11020	Física / Matemàtiques (simultaneïtat) (Barcelona)	UB	Barcelona	Barcelona	Double Degree	Science and Technology	13.170
31112	Enginyeria en Tecnologies Aeroespacials, Grau en + Màster Universitari en Enginyeria Aeronàutica (PARS: Enginyer/a Aeronàutic/a)	UPC	Terrassa	Barcelona	Engineering and Architecture	Science and Technology	12.936
11016	Enginyeria Informàtica / Matemàtiques (simultaneïtat) (Barcelona)	UB	Barcelona	Barcelona	Double Degree	Science and Technology	12.898
11043	Medicina (Campus Clínic) (Barcelona)	UB	Barcelona	Barcelona	Health Sciences	Science and Technology	12.850
11059	Enginyeria Biomèdica (Barcelona)	UB	Barcelona	Barcelona	Engineering and Architecture	Science and Technology	12.740
21145	Dret / Relacions Internacionals (simultaneïtat) (Cerdanyola del Vallès)	UAB	Cerdanyola del Vallès	Barcelona	Double Degree	Social Sciences	12.714
41058	Medicina (Barcelona)	UPF	Barcelona	Barcelona	Health Sciences	Science and Technology	12.650
11083	Medicina (Campus Bellvitge) (L'Hospitalet de Llobregat)	UB	L'Hospitalet de Llobregat	Barcelona	Health Sciences	Science and Technology	12.650
31040	Matemàtiques (Barcelona)	UPC	Barcelona	Barcelona	Sciences	Science and Technology	12.644

In order to carry out the cleaning and structuring process, I did the following:

1. Extracting Location from Degree Name

All degree names in the dataset ended with their location in parentheses, for example, Administració i Direcció d'Empreses (Barcelona). My goal was to systematically extract just the place name (e.g., Barcelona).

To do this, I used two Excel formulas:

- TEXTAFTER(A1, "(", -1): This extracts all text after the last opening parenthesis in the string from cell A1. For the example above, it would yield Barcelona).
- TEXTBEFORE(..., ")"): Applying this to the previous result, it pulls everything before the closing parenthesis, giving just Barcelona.

Combined into one formula: =TEXTBEFORE(TEXTAFTER(A1,"(",-1), ")").

fx =TEXTBEFORE(TEXTAFTER(C6;"(";-1);")")

C	D	E
Degree	University	Location
Física / Matemàtiques (simultaneïtat) (Cerdanyola del Vallès)	UAB	Cerdanyola del Vallès
Física / Matemàtiques (simultaneïtat) (Barcelona)	UB	Barcelona
Enginyeria en Tecnologies Aeroespacials, Grau en + Màster Universitari en Enginyeria Aeronàutica (PARS: Enginyer/a Aeronàutic/a)	UPC	Terrassa

2. Creating and Filling the Region Column

Once I extracted the city names, I wanted to have a broader region classification, in this case, the **Catalonia provinces**: Barcelona, Tarragona, Girona and Lleida. This classification will be used to the final API so that students living in Girona, for example, can filter Girona-based universities.

In order to do so, I created an Excel tab called "Data_Cleaning" where I created a mapping table with two columns: the first one is the city values where I extracted these by executing the "Unique" Excel formula (**45 locations**) of the Column "Location", and later I allocated its specific province.

After that, I added the Region column by executing a **lookup** to fill the region column automatically for all rows. The photo on the right shows just some of the regions.

Unique Regions from Table 1 - Column E	Province
Mataró	Barcelona
Barcelona	Barcelona
Cerdanyola del Vallès	Barcelona
Girona	Girona
Igualada	Barcelona
Lleida	Lleida
Manresa	Barcelona
Reus	Tarragona
Tortosa	Tarragona
Vic	Barcelona
Tarragona / Reus	Tarragona
Reus / Barcelona	Tarragona
Tortosa / Barcelona	Tarragona
Tarragona	Tarragona
Sant Cugat del Vallès	Barcelona
Salt	Girona

3. Adding the Field + High School Specialization columns

To enable more advanced filtering and personalized recommendations, I enriched the dataset by adding two key columns: **Field** (field of the study of the bachelor's degree) and **High School Specialization** (the Bachillerato track that typically grants access to each degree). This allows the platform to let students filter degrees by either their current specialization, their field of interest, or both. For example, a student from a Science and Technology background could still explore management or law degrees if those are accessible or relevant.

For the **Field column**, my goal was to assign every degree to a **broader academic area** such as "Health Sciences," "Engineering and Architecture," or, for combined programs, "Double Degree" (so I could provide a filter specifically for dual programs). To automate this field column, I used an LLM tool in which I prompted to extract relevant keywords from each degree title and map them to the correct academic area. It's also located in the "Data_Cleaning" tab. The photo on the right shows some of these keywords. For example, any degree containing "Enginyeria" was automatically mapped to "Engineering and Architecture." Wherever the LLM or pattern-based lookup didn't retrieve a clear result, such as less common or ambiguous degrees, I assigned the field manually to ensure accuracy. The following is a non-exhaustive list of keywords:

Keyword	Field
(simultaneïtat)	Double Degree
Treball	Social Sciences and Law
Enginyeria	Engineering and Architecture
Arquitectura	Engineering and Architecture
Química	Sciences
Biologia	Sciences
Ciències Ambientals	Sciences
Matemàtiques	Sciences
Física	Sciences
Empresa	Social Sciences and Law
Administració	Social Sciences and Law
Economia	Social Sciences and Law
Dret	Social Sciences and Law
Turisme	Social Sciences and Law

For the **High School Specialization** column, I followed a similar approach. I built a mapping table that links each field to its most common Bachillerato specialization.

Batxillerat	Batxillerat
Double Degree	Manual
Social Sciences and Law	Social Sciences
Engineering and Architecture	Science and Technology
Sciences	Science and Technology
Health Sciences	Science and Technology
Arts and Humanities	Arts and Humanities

4. Final Cleaning & Export

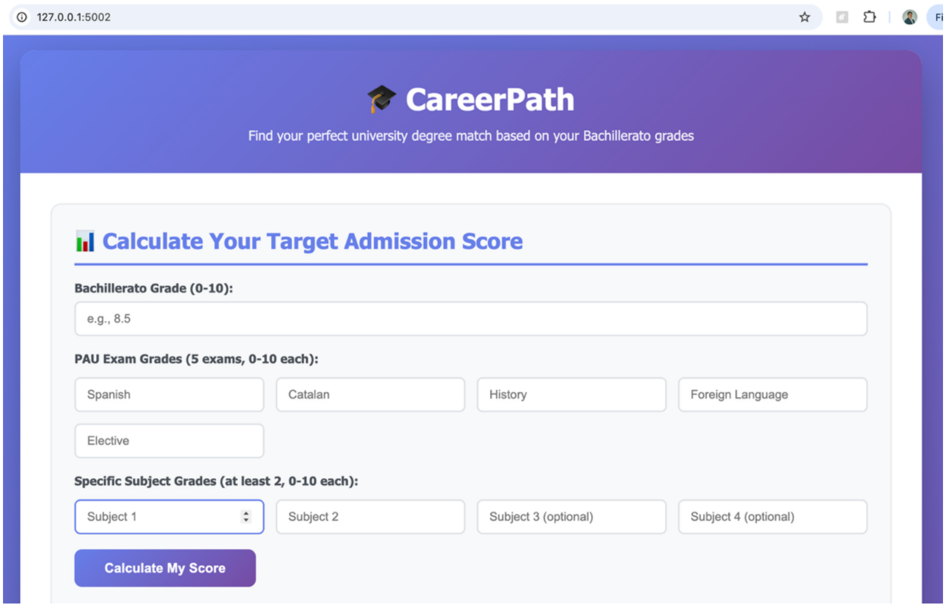
As a final step, I made sure all categorical values were consistent (for example, always spelling "Barcelona" the same way), and ensured that all necessary fields were filled in. With all columns complete and clean, I then converted the Excel table into a JSON file, so it's much convenient for direct use in the API with the HTML backend.

3. ENDPOINTS and OUTPUT

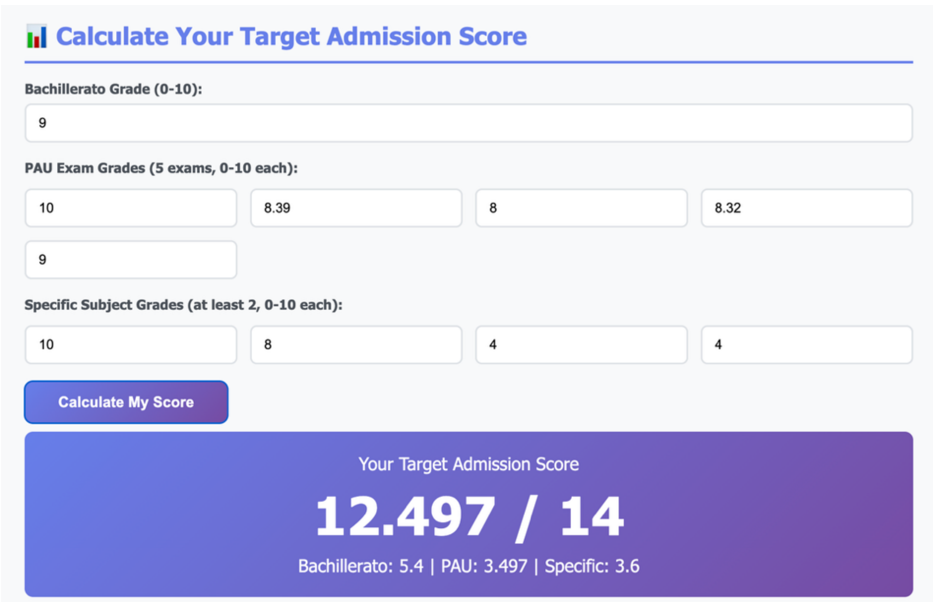
For this project I used 5 endpoints:

- **/api/v1/students/calculate-score (POST)**: Calculates the student's university admission score (out of 14) based on Bachillerato, PAU exams, and specific subject marks.
- **/api/v1/students/recommendations (POST):** Returns a ranked list of university degrees that best match the student's score, specialization, field of interest, location, and double degree preference.
- **/api/v1/statistics/overview (GET):** Presents summary statistics including score distribution, fields, specializations, and universities in the dataset.
- **/api/v1/degrees/search (GET):** Enables filtering degree programs by field, score range, university, location, specialization, and double degree status, without requiring an admission score.
- **/api/v1/degrees/<code> (GET):** Retrieves complete details for a specific degree program by its unique code.

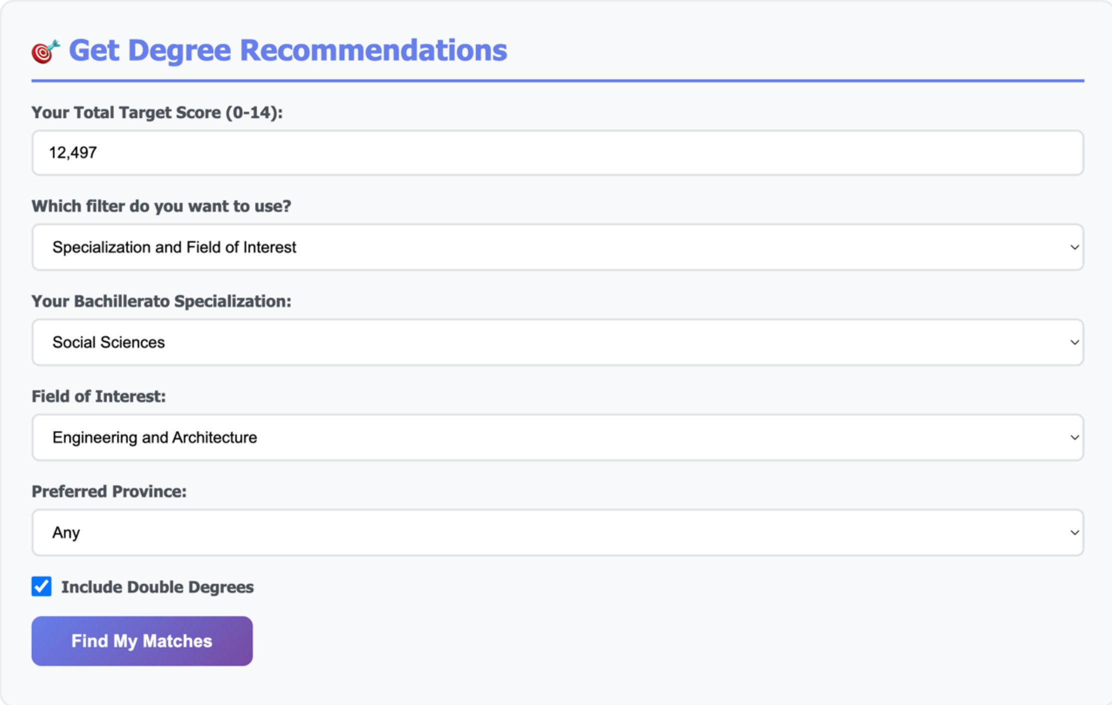
This is the initial view of the interface, where students can enter their Bachillerato and PAU grades, as well as specific subject scores, to begin the recommendation process.



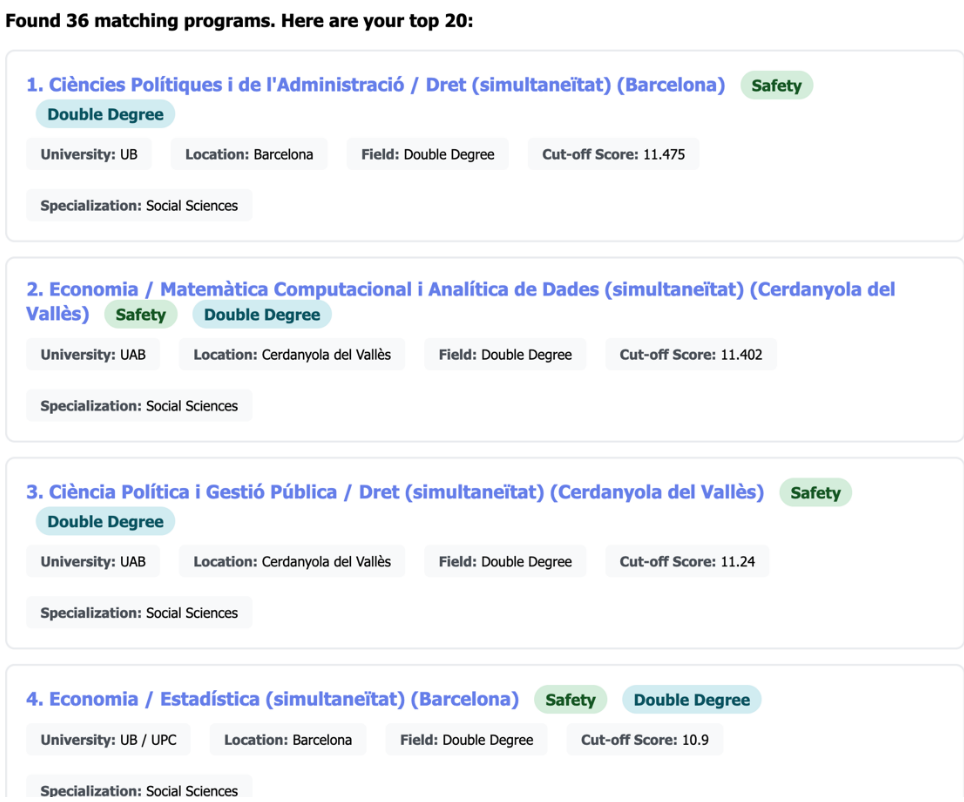
Once grades are entered and the score is calculated, the interface displays the student's target admission score. This output corresponds to the functionality provided by the first API endpoint (calculate-score).



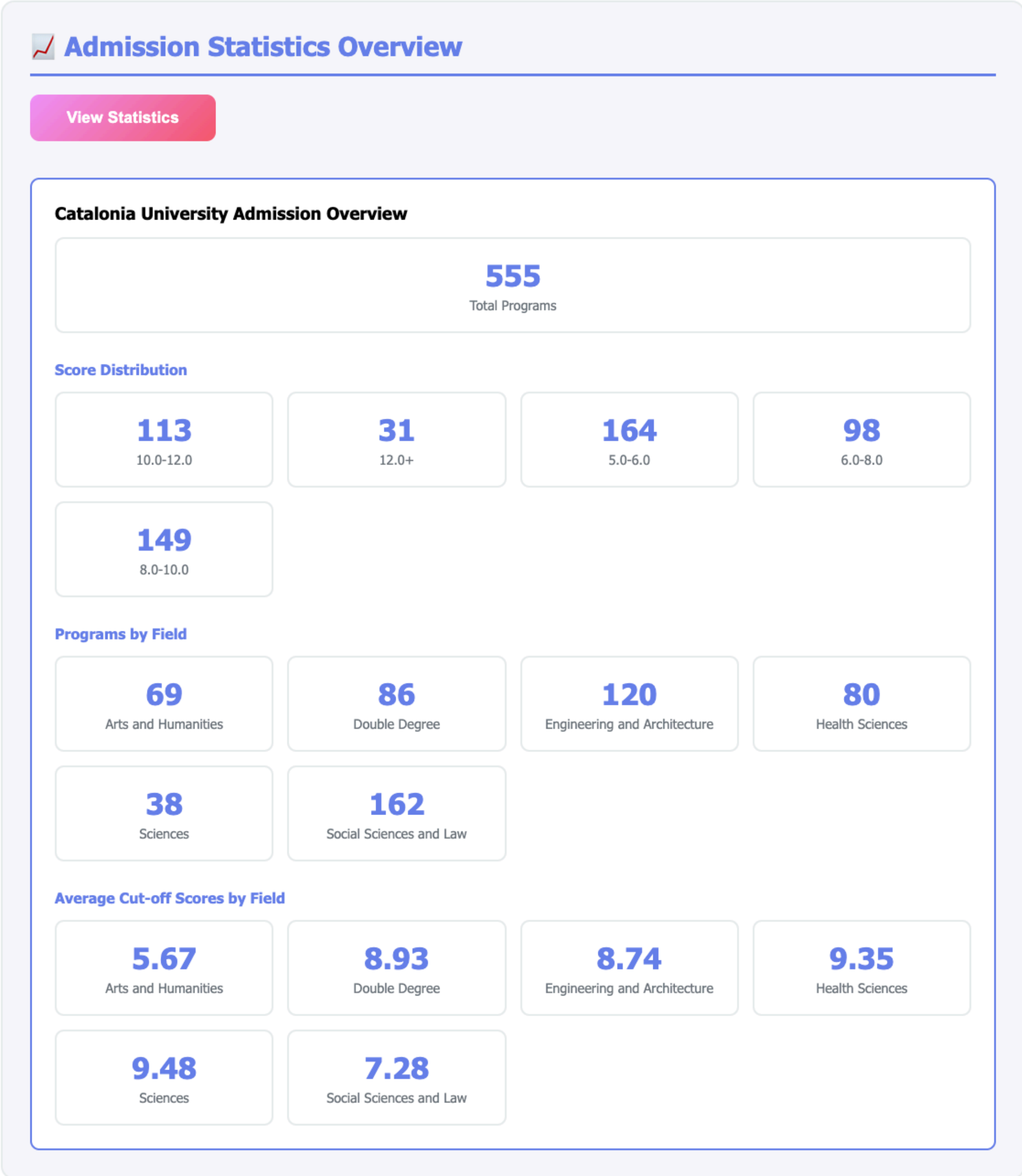
Next, students can set various filters according to their **field preferences and high school specialization** as well as other variables such as field of study, location, or double degree options. This process is linked to the second API endpoint (recommendations).



Once you click Find My Matches, it will show the student the **Top 20 options**. The example shown below is a sample subset for demonstration purposes.



At the bottom of the interface, you can access an **overview of statistics**, which summarizes information about all 555 bachelor programs in Catalonia.



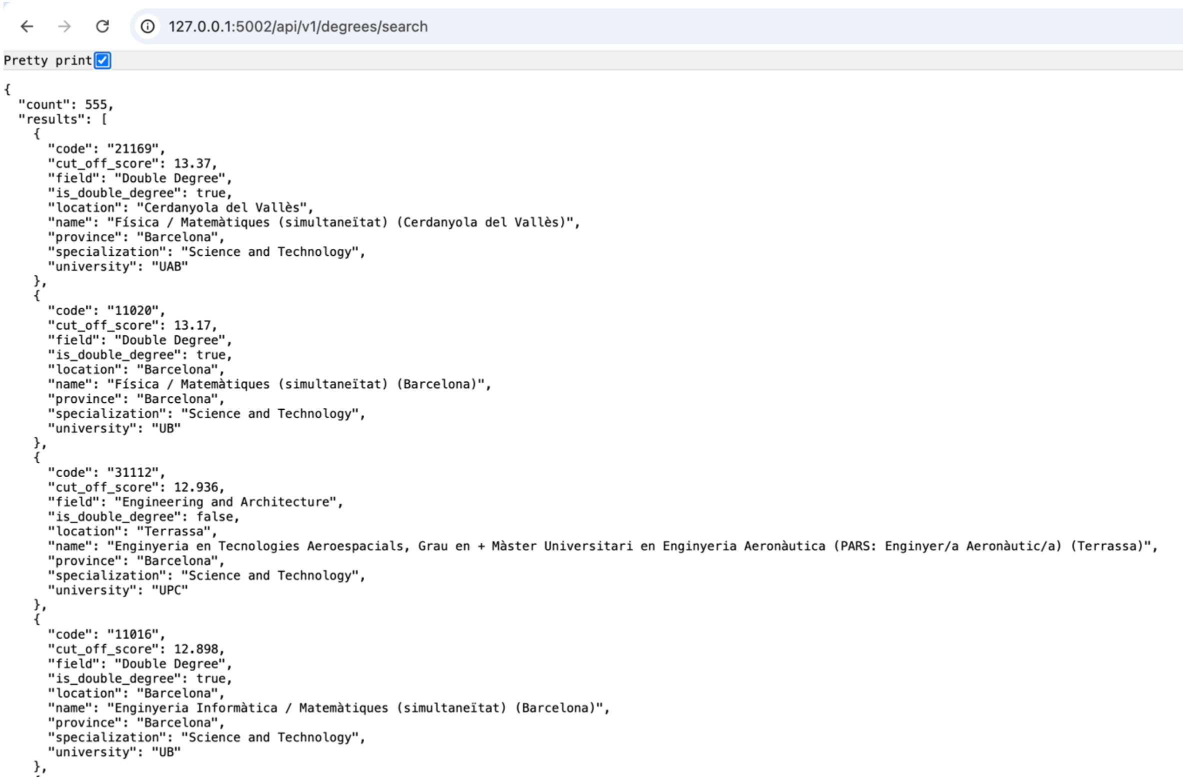
Alternatively, you can also retrieve the third endpoint like this:

127.0.0.1:5002/api/v1/statistics/overview

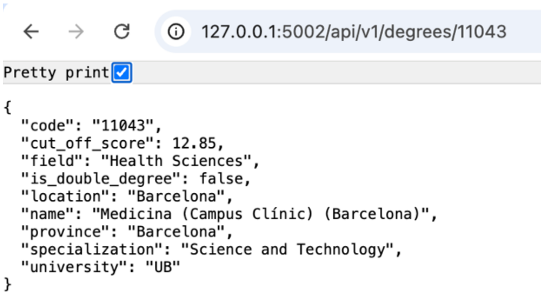
Pretty print


```
{
  "average_scores_by_field": {
    "Arts and Humanities": 5.67,
    "Double Degree": 8.93,
    "Engineering and Architecture": 8.74,
    "Health Sciences": 9.35,
    "Sciences": 9.48,
    "Social Sciences and Law": 7.28
  },
  "programs_by_field": {
    "Arts and Humanities": 69,
    "Double Degree": 86,
    "Engineering and Architecture": 120,
    "Health Sciences": 80,
    "Sciences": 38,
    "Social Sciences and Law": 162
  },
  "programs_by_specialization": {
    "Arts and Humanities": 86,
    "Science and Technology": 270,
    "Social Sciences": 199
  },
  "programs_by_university": {
    "UAB": 111,
    "UB": 90,
    "UB / UPC": 2,
    "UPC": 67,
    "UPC / UB /\nUAB / UPF": 1,
    "UPC / UPF": 1,
    "UPF": 47,
    "UPF / UAB": 1,
    "URV": 66,
    "URV / UOC": 1,
    "UVic-UCC": 39,
    "UVic-UCC /\nUB": 1,
    "UdG": 70,
    "UdG / UB": 2,
    "UdL": 56
  },
  "score_distribution": {
    "10.0-12.0": 113,
    "12.0+": 31,
    "5.0-6.0": 164,
    "6.0-8.0": 98,
    "8.0-10.0": 149
  },
  "total_programs": 555
}
```

To access more advanced options, you can use the forth endpoint, which supports searching and filtering degrees using custom parameters directly in the API request (related to the third endpoint).



If you know the unique code for a particular degree, you can retrieve its full details using the fourth endpoint by specifying the official code of the bachelor's degree.



4. Conclusion

This project aims to provide a personalized advisory application for high school students in Catalonia, making it possible for them to match their predicted Bachillerato and PAU grades to university degree programs in a simple and interactive way. The current version offers a practical tool for (i) calculating admission scores, (ii) instantly retrieving the top 20 bachelor programs tailored to each student's academic profile and (iii) retrieving general statistics about cut-off scores in Catalonia, with a core focus on usability and clarity.

There remain some opportunities for further improvement. For instance, I'd intend to integrate the more advanced endpoints, such as program search (/api/v1/degrees/search) and individual degree details (/api/v1/degrees/<code>), into the HTML frontend. This would helping students to filter programs with greater flexibility or access comprehensive information on specific degrees.